

Engr.Sheraz umer

+92-3028150399

shirazisahb134@gmail.com

linkedin.com/in/sheraz-umer-4691a9201

• Lahore-54000,Punjab,Pakistan

OBJECTIVE

Job-oriented Fresh graduate in Electrical Engineering with specialization in **Power System** with hands-on experience in **Switchgears(LV/HV Devices)** designing **Power Converters** and taking **Simulation** results through a comprehensive Final Year Project. Skilled in **Embedded Systems, Control System,Power System(Generation,Transmission,Utilization)**,have skilled in C/C++(Matlab,Labview) and Python. Proficient in **Microcontroller** programming(FPGAs,AurdinoUNO,Esp32/86) and designing Electrical/Electronics Circuits in **PCB,XeroBoard** .Adept at utilizing **AutoCAD,Solidworks** and **PlanSwift** softwares to analysis modern days problems and to develop innovative solutions of real-world.

WORK EXPERIENCE

Cost Estimation Engineer, G4S Estimation Company

June 2024 – Present

Lahore, Pakistan

- Prepared detailed cost estimates and takeoffs for material, labor, and overhead costs.
- Interpreted and analyzed project plans and specifications for accurate cost evaluations.
- Estimated costs for HVAC systems, integrating HVAC components into project budgets.
- Managed project timelines and budgets, ensuring timely completion and adherence to deadlines.
- Contributed to competitive bidding by preparing proposals and cost breakdowns for tenders.
- Worked effectively with engineers, architects, and clients to achieve project goals.

Intern, Roulite Techno

December 2023 – February 2024

Sukkur,Pakistan

- Assisted in the installation and commissioning of photovoltaic (PV) solar systems, including module mounting and inverter setup.
- Utilized monitoring tools to track system performance, analyze data, and identify efficiency improvements.
- Developed skills in diagnosing and resolving issues related to solar panels, inverters, and electrical wiring.
- Adhered to safety standards and regulations specific to solar installations, ensuring safe work practices and procedures.

Executive Member, Engineering Society S-IBA

December 2023 – February 2024

Sukkur,Pakistan

- Organized and coordinated engineering-related events, workshops, and seminars.
- Assisted in the planning and execution of student engagement activities and technical sessions.
- Collaborated with faculty and industry professionals to facilitate guest lectures and networking opportunities.
- Contributed to the development of promotional materials and communications for society events.
- Supported the team in managing resources and ensuring smooth execution of society initiatives.

Intern, OGDCL Dakhni Gas Plant

September 2023 –November 2023

Attock, Pakistan

- Developed a comprehensive understanding of gas turbine operations and combined cycle power plant mechanisms.

- Observed and analyzed the responsibilities of Electrical Engineers in LV/HV switchgear rooms.
- Acquired hands-on experience with HMI, SCADA, PLC control systems for monitoring and automation.

Intern, SEPCO 132KV Grid Station
Sukkur, Pakistan

June 2023 – August 2023

- Managed 132KV power transmission systems, including transformers and circuit breakers.
- Utilized SCADA for real-time monitoring and system control.
- Adhered to high-voltage safety standards and Safety protocols.
- Diagnosed and resolved electrical faults and malfunctions.
- Coordinated with maintenance and operations teams, and documented technical processes.

EDUCATION



Sukkur IBA

(BE) Electrical Engineering **2020-2024**

Sep 2020 - May 2024 | **CGPA = 3.34 out of 4**

| **Grades = A 85%**



Barret Hodgson International College TTS

(HSSC) FSC Pre-Engineering **2017-2019**

Aug 2017 - Aug 2019 | **Grade = A 80%**

| **Marks 852/1100**



Allied School Sargodha Road Campus

(SSC) Matric Science **2014-2016**

Feb 2014 - May 2016 | **Grade = A+ 90%**

| **Marks 992/1100**

KEY SKILLS

- Designing Models and Simulating Results Power Converters (Expertise in MMC Converters)
- Embedded System and Microcontroller Programming (FPGAs ,ESP 32/8266, Arduino Uno)
- Comprehensive Knowledge on Programmable Logic Controllers (PLCs) & SCADA.
- Designing Digital Logic Design (DLD,DSD),Half Adder,Full Adder,Implementing Logic Gates.
- Making SLDs ,Load/Power Flow Analysis & Steady State Analysis.
- Electrical Machines (Transformers and DC/BLDC Motors).
- Printed Circuit Design PCB ,double/single layers ,Preparing BOM & Gerber File.(Altium,EasyEDA).
- Project Management and Report Writing through MS WORD,LATEX.
- Thermodynamics (Comprehensive knowledge on Boilers and Steam Turbines).
- Instrumentation and Measurement (Sensing and Actuating Devices).
- ETAP & PLECS, Power Factory, PSCAD, PSS/E, OPAL-RT Power Hardware in Loop (PHIL) simulator.
- Taking Takeoffs, Linears, Counts and Estimates and Electrical CSI Divisions through PLANSWIFT.
- NI LabView, MATLAB/Simulink, Multisim, SolidWorks, Planswift, Autodesk, Maxwell Ansys.

PROJECTS

- **Advanced Hardware-in-Loop Setup and Digital Control for Power Converter Optimization**
Final Year Project FYP *Sep2023 – June 2024*
Conducted comparative analysis of PSC-SPWM and NLM modulation techniques on MMC using MATLAB/Simulink. Developed a three-phase motor control system on GPIC for real-time testing, enhancing the teaching and research of steady-state and dynamic behavior of power converters.

- **Efficient Hardware-in-Loop Setup and Digital Control Technique** *Sep 2023 – Oct 2023*
Developed a Hardware-in-Loop setup for power systems to enhance learning and bridge theoretical and practical gaps.
- **20 kW Grid-Connected PV System Analysis** *Sep 2023 – Nov 2023*
Analyzed performance metrics using PVSyst for a 20 kW photovoltaic system.
- **Smart Switches Using IoT** *Feb 2022 – Dec 2022*
Created smart switches with IoT for remote control and monitoring, integrating sensors and cloud data visualization.
- **Analog Clock Design Using MATLAB** *Sep 2022 – Dec 2022*
Designed an analog clock with hour, minute, and second hands using MATLAB and trigonometry.
- **3-Φ Induction Motor Optimization** *Sep 2023 – Jun 2024*
Improved efficiency of a three-phase induction motor for irrigation pumps.
- **PV System Simulation** *Jun 2023 – Jul 2023*
Simulated a 20 kW grid-connected photovoltaic system using PVSyst.

WORKSHOPS

Technical Presentation and Communication Skills

Jan 2023 |

Sukkur IBA

- One-day workshop on effective technical presentation and communication skills.

3D Design and Fabrication Laboratory (FabLab)

Oct 2022 |

Sukkur IBA

- Three-day workshop on learning, innovation, and creativity in FabLab.

Integrated Circuit (IC) Design

Dec 2023 |

NED,FAST,Sukkur IBA

- Two-day online workshop on IC design technologies and 10XEngineer,Aql semiconductor collaboration.

AWARDS & HONOURS

- **PEC Funded Final Year Project** *Jan 2024 |* *Pakistan Engineering Council PEC*
Received funding for the final year project on power converters optimization.
- **PM Laptop Scheme Selection** *Nov 2023 |* *Government of Pakistan GOP*
Awarded a laptop under the Prime Minister's Laptop Scheme.
- **OGDCL NTHP Scholarship** *Aug 2019 |* *Sukkur IBA University*
Secured a fully funded scholarship for the Bachelor program at Sukkur IBA University.

LANGUAGES

- **English:** Advanced proficiency in reading, speaking, listening, and writing.
- **Urdu:** Advanced proficiency in reading, speaking, listening, and writing.
- **Punjabi:** Native language. Could be provide upon request.

REFERENCES

Could be provide upon request.